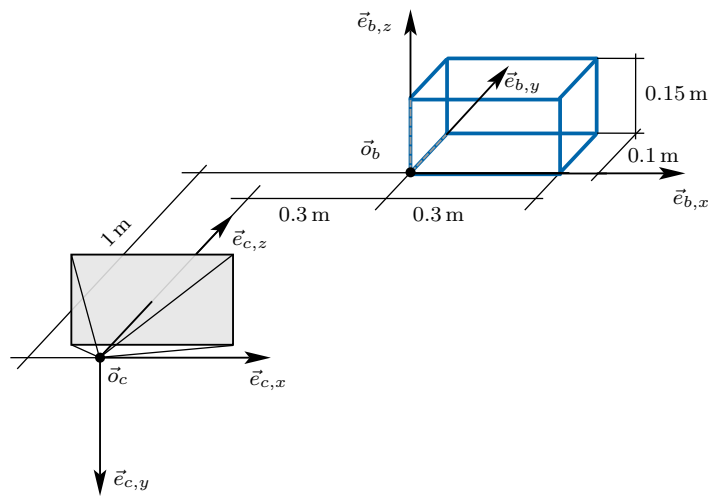


12 Laboratory – Camera Simulation

Learning goals

- you can properly choose the focal length of a camera
- you can intuitively express the pose between object and camera
- you can “simulate” the projected image of a 3D-object

Introduction



The sensor chip of the camera has the dimensions 8×6 mm and a resolution of 4000×3000 pixels, its center lies on the position $\vec{o}_c$ and its long edge is parallel to the x -axis

The projection of object points expressed in b -coordinates to image coordinates is described with the formula from the lecture:

$$\underbrace{\begin{bmatrix} b_x^c p_z \\ b_y^c p_z \\ p_z \end{bmatrix}}_{\check{b}^c} = \underbrace{\begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}}_{K_c} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \underbrace{\begin{bmatrix} R_b^c & t_{cb}^c \\ [0, 0, 0] & 1 \end{bmatrix}}_{T_b^c} \cdot \underbrace{\begin{bmatrix} p^b \\ 1 \end{bmatrix}}_{\check{p}^b} \quad (1)$$

Exercise 1. Focal Length and Pose

- a) Determine the focal length F of the camera such that the sensor chip just captures the right border of the object.
- b) Determine the focal lengths f_x and f_y .
- c) Determine the image center coordinates c_x and c_y for an ideal camera, i.e. with the optical axis pointing exactly through the center of the sensor chip.
- d) Express the relative pose $\mathbf{T}_b^c \in \mathbb{R}^{4 \times 4}$ in numbers.

Exercise 2. Image Simulation

- a) Complete the program to compute the 8 corner points of the object. A drawing method is already prepared in the scaffold.
- b) Now, rotate the object around its z-axis $\vec{e}_{b,z}$ by some angle α and let the image be visualized again by your script. Now we must distinguish between (body) b-coordinates and rotated body coordinates which we shortly call r -coordinates. The resulting matrix can be written as

$$\mathbf{R}_b^r(\alpha) = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

To keep our transforms layed out clearly, we rephrase (1):

$$\underbrace{\begin{bmatrix} b_x^c p_z \\ b_y^c p_z \\ p_z \end{bmatrix}}_{\check{\mathbf{b}}^c} = \underbrace{\begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}_c} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \underbrace{\begin{bmatrix} \mathbf{R}_r^c & \mathbf{t}_{cr}^c \\ [0, 0, 0] & 1 \end{bmatrix}}_{\mathbf{T}_r^c} \cdot \underbrace{\begin{bmatrix} \mathbf{R}_b^r(\alpha) & \mathbf{t}_{rb}^r \\ [0, 0, 0] & 1 \end{bmatrix}}_{\mathbf{T}_b^r} \cdot \underbrace{\begin{bmatrix} \mathbf{p}^b \\ 1 \end{bmatrix}}_{\check{\mathbf{p}}^b},$$

with $\mathbf{t}_{rb}^r = [0, 0, 0]^T$.